```
)

print("=== Individual-level: Which early behaviors differ the most (COUNTED) ===")
display(indiv_count_df)
```

```
=== Individual-level: Which early behaviors differ the most (COUNTED) ===
```

code	mean_unsuccessful	mean_successful	percent_diff
Idea Management	0.706510	0.891249	26.148165
Integration Practices	0.089652	0.107044	19.399459
Information Seeking	0.436641	0.497858	14.019849
Relational Climate	0.419338	0.464851	10.853454
Knowledge Sharing	1.392875	1.463399	5.063151
Coordination and Decision Practices	0.331531	0.336166	1.397945
Participation Dynamics	0.222752	0.222367	-0.172778
Evaluation Practices	0.308715	0.301198	-2.434853

Slide 10 - Original Attempt

Went through all of the code chunks (it is shown in the `linkography-code-annotation.pdf` in my repository under the `Dec-2025-status-report` folder).

Below is a rundown explanation of how I got through that attempt.

I started by treating the annotation codes as basic behavioral building blocks, rather than the labels they were in the codebook. This is because I kept relaying back to the idea of team formation, rather team funding as the finalized output -- that's what was on my mind at the time cause everything keeps tying back to that. Cause I was trying to describe at the certain parts, what gets a team funded?

So then trying to piece those parts together because there could be up to three codes with one utterance, I was trying to make it a bit simpler on myself. As I also knew the annotation codes wouldn't be able to tell the story by themselves - I also ended up using the respective transcripts to end up checking it as well. For this, I ended up grouping the original annotaiton codes by the cognitive function they serve in a team's reasoning process, rather than focusing on the duration or frequency a specific annotation code shows up.

That led to this idea of something like the following (just to keep my head intact, not saying all meetings followed this structure to a tee):

Seeding/Expansion --> Stress-testing --> Synthesis --> Commitment

I had grouped `Knowledge Sharing` and `Information Seeking` under a `Expansion/Seeding` category because both of these behaviors introduce or clarify infomration that expands the set of possible ideas under consideration. `Knowledge Sharing` ends up bringing in content into the conversation; `Information Seeking` deals with missing information or clarification.

I had grouped `Evaluation Practices` on its own under a `Stress-testing` category because it does not expand the idea space, but rather compares, critiques or constrains the existing ideas. This all represents `stress-testing`, which alternatives are weighed against feasibility, overall group goals, or other possible constraints. This had often marked a turning point between the exploration and convergence.

I had grouped `Integration Practices` as a `synthesis` category because it involves the combining, reconciling, or reframing multiple ideas into a more robust structure. This actively constructs something new from exisiting ideas.

I had finally grouped `Coordination and Decision Practices` as a `commitment` behavior. This effectively signals alignment and forward movement; not necessarily an idea exploration. This may include choosing a direction, assigning tasks, defining next steps, formalizing decisions. Not all of a scientific teams' interactions would show this behavior, but this behavior was the last "part" that still needed to be included.

`Idea Management` was given its own category of `IDEA_MGNT` because it captures the meta-behavior about how ideas are tracked, referenced, or organized over time. This also may include behaviors such as labeling ideas, revisiting earlier suggestions, considering alternative ideas, transitioning between ideas, or even just the continuity throughout the meeting -- keeping the structure together.

For this, `Relational Climate` was never filtered into one of these categories, or given its own, because it was more so about the emotional aspect of these scientific team meetings, cause I again, kept tying all of this to the proposal outcomes.

After I had defined all of these higher-level behavioral, I then incorporated all of this back into the temporal third segments (beginning, middle, end). The goal was to test whether these grouped behaviors shift over time in ways consistent with convergence. Because of the large amount of meetings and knowing they all would be different in terms of structure, I had focused on how these grouped behaviors transition into one another. This then allowed me to find out which behavioral states, trying to keep in-tact an idea-state flow rather than isolating the individual annotation codes. For this I treated `IDEA_MGMT` as a way to support the transitions without implying convergence (also to keep true to the codebook), to seeing if a relationship with these behaviors hold. This is before tying it to outcomes such as team formation or funding.

Once all these transitions were figured out, I had to figure out whether a `synthesis` or `commitment/closure` reflected genuine idea-level convergence or saying it was the end of a meeting. For these two behaviors (`synthesis` and `commitment/closure`, which involves the behaviors these are associated with in the codebook - see earlier section) I had figured out that these coordination and decision-like behaviors often increase near the end of the meetings due to the facilitation, reporting, or time management - not necessarily due to substantive agreement. I ended up running a false-positive diagnostic analysis while auditing the transcript for reference and using a lexical pattern detection. I then inspected these segments in context to distinguish when the meeting actually finished (because they ran out of time) compared to a genuine convergence, when people were wrapping up on an idea.

This then led me to figuring out that the narrowing does not rely on the behavioral annotations. This is what led me to figure out whether the diversity of ideas under discussion decreases over time. This led me to using an entropy to measure this conceptual spread. This motivation was because that the genuine convergence (e.g. a best idea, correct idea, final decision, successful outcome, etc.) should not only be reflected in how teams behave, but also to the extend the ideas are actually being considered. I then computed the entropy separately for the first-third and last-third of each meeting. For that, I defined a pruning score as the change in entropy over time -- negative values indicate narrowing of an idea space.

After this part was run, it was figured out that some of the meetings had strong `synthesis` and `commitment/closure` behaviors without a reduction in the entropy. Others showed conceptual narrowing without clear `commitment/closure` behavior. For this part I ended up checking against the transcript to figure out the exact cases and words spoken by the researchers.

Addressing this limitation, this holds because teams can exhibit `synthesis` and coordination behaviors without adhering to a specific idea, and they can narrow their idea space without signaling commitment through decision-oriented behaviors. This therefore shows that these annnotations and transition dynamics can help figuring out where convergence of an idea may be occuring, but insufficient on what the team has converged on. Some sort of semantic tracking would be required to confirm genuine ideal-level closure.

So then given all of this, **to assess decision closure**, I computed late-stage gains in Commitment/Closure relative to early segments, and examined whether these gains occurred after synthesis rather than in isolation. However, because commitment-like behaviors often occur near the end of meetings due to facilitation or reporting, I implemented a false-positive diagnostic using lexical pattern detection. Specifically, utterances containing facilitation or report-out language (e.g., time management, slide preparation, reporting instructions) were flagged via regex-based matching and audited in the transcript. Commitment or synthesis signals overlapping with these patterns were treated as potential structural closure rather than genuine idea-level convergence.

To assess narrowing independently of behavioral annotations, I computed idea-space entropy for the first and last thirds of each meeting, treating the distribution of idea states as a proxy for conceptual spread. A pruning score was defined as the change in entropy over time, with negative values indicating narrowing of the idea space. This provided a code-agnostic measure of whether teams reduced the range of ideas under active discussion.

Finally, I compared behavioral convergence signals (`synthesis` and `commitment` gains) with entropy-based narrowing. Meetings where these signals aligned were treated as strong candidates for convergence, while mismatches were examined via transcript inspection. These mismatches revealed that behavioral convergence and conceptual narrowing do not always co-occur, motivating the conclusion that behavioral annotations can localize where convergence may be occurring, but are insufficient to determine what idea a team has converged on without additional semantic tracking.